

Vital Sleep Patch:

Transforming Pediatric Sleep Apnea Diagnosis through Home-Based Testing

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★ Pediatric Obstructive Sleep Apnea (OSA) severely impacts children's life and health.

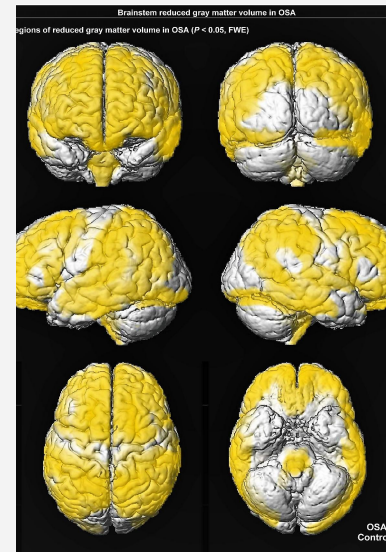
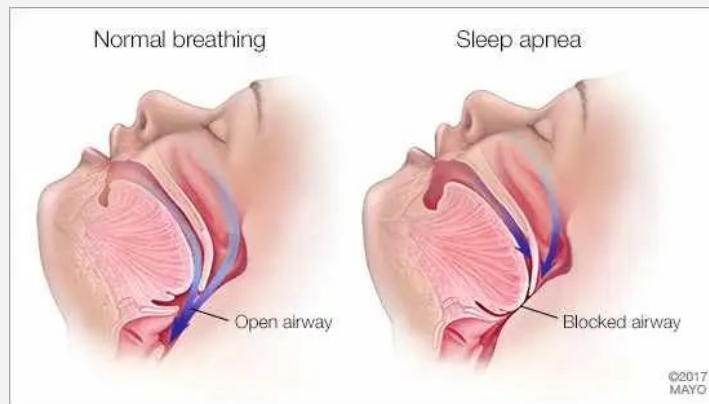


Sumit Bhargava, MD

Clinical Professor, Pediatrics

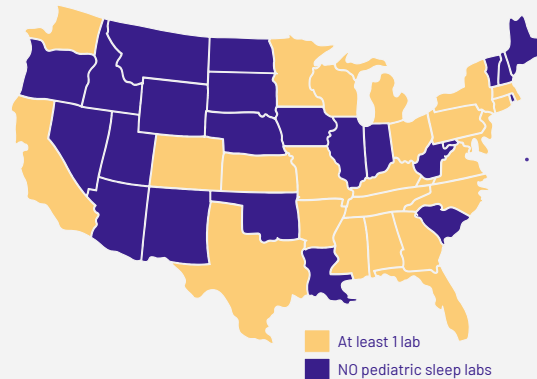
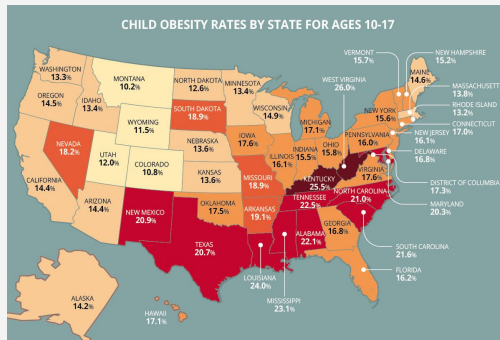
Pulmonology & Pulmonary Medicine

Practices At: Stanford Medicine Children's Health



Effects span from **metabolic dysfunction** and **behavioral problems** to **decreased test scores** and even **structural changes to the brain**.

Millions of kids need treatment, but the diagnostic pipeline is broken.



~1 in 15 kids

in the US have obstructive sleep apnea (OSA), yet **up to 90% remain undiagnosed**

[[State of Childhood Obesity](#), [AAPA](#), [Osteopathic](#)]

~1 in 5 kids

in the US are obese: nearly **15 million children** between ages 2 and 19. Up to 60% of them have sleep apnea.

[[CDC](#)]

Half the states in the U.S.

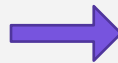
have no accredited pediatric sleep labs

[[AASM Sleep Education](#)]

The Vital Sleep Patch offers a paradigm shift:

Problem:

In-clinic polysomnography is failing to address pediatric sleep apnea nationwide



Solution:

A send-home wearable sensor patch and AI-based diagnostic pipeline.

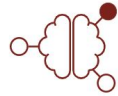


This technology is ready to leave Stanford research labs and enter the real world.

STANFORD
MUSSALLEM CENTER FOR
BIODESIGN



Over the past 5 years, we've worked with Stanford Biodesign, the Coulter Translational Research Program, Stanford Children's Health, and Stanford Bioengineering to ideate, prototype, and validate our solution.



STANFORD UNIVERSITY
NEURAL INTERACTION LAB

- Unique pediatric focus throughout engineering & design
- IP strategically positioned to avoid competition
- Papers & patents being filed for sensing hardware + AI algorithms

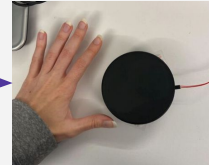
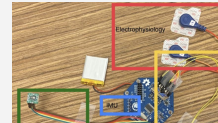
1. Approach Validation with n = 6 subjects



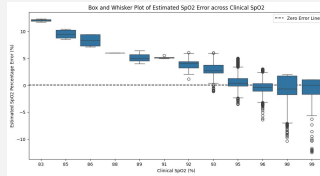
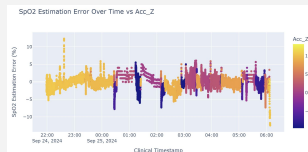
Patient	PSG AHI	HSAT AHI	PSG Severity	HSAT Severity
1	140.3	65.0	Severe	Severe
2	73.3	11.0	Severe	Severe
3	5.8	5.0	Moderate	Moderate
4	3.0	2.9	Mild	Mild
5	7.5	1.7	Moderate	Mild
6	14.5	18.7	Severe	Severe



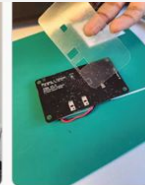
2. Prototype Progression



4. AI Algorithm Validation with n = 10 subjects



3. Sensor Validation with n = 20 subjects



The Dream Team

The Picker

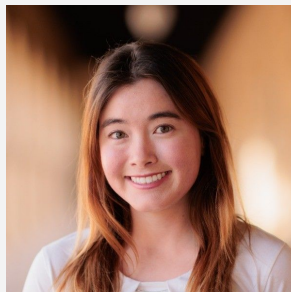


Brian Boitnott
([LinkedIn](#))

PhD Candidate in
Biomedical Physics

Previously:
MIT Lincoln Laboratory
(Intelligence, Surveillance,
and Reconnaissance)

The Designer



Alisa Hathaway
([LinkedIn](#))

PhD Candidate in Electrical
Engineering + Masters in EE

Previously:
MIT Computation &
Cognition, Northrop
Grumman, NextSense

The Hacker



Syamantak Payra
([LinkedIn](#))

PhD Candidate in Electrical
Engineering + Masters in MS&E

Previously:
MIT EE+CS, SpaceX, Apple,
McKinsey, UnitedHealthcare

The Hustler



Soham Saha
([LinkedIn](#))

Second year MBA
Student

Previously:
IIT Engineering, Google,
Merck, Zyla Health

Team Advisors



Sumit Bhargava, MD

Medical Director of the

Stanford Children's Hospital Pediatric Sleep Laboratory

Expert on pediatric sleep apnea



Todd Coleman, PhD

Associate Professor of Bioengineering at Stanford

Fellow of the American Institute for Medical and
Biological Engineering

Expert on wearable sensing systems



Key Partners  Academic medical centers: clinical validation, research collaboration, access to patient cohorts Private practices: access to last-mile patient cohorts and feedback on device usability Sleep medicine clinics: specialized expertise in sleep disorders Medical hardware manufacturers: Device production and scaling expertise Cloud/Remote health companies: HIPAA compliant data storage and remote health infrastructure	Key Activities  Validation: finish testing against gold standard (hospital study) Algorithm: refine AI-accelerated diagnostic capability FDA Pre-Submission: complete paperwork for compliance Manufacture first “production” devices Key Resources  Sensing hardware IP & engineering experience Diagnostic algorithm IP and software expertise Clinical translational experience Product management expertise	Value Propositions  Patients: - Faster and more streamlined diagnostic of pediatric OSA, allowing for earlier treatment - improves health and life-expectancy and increases quality of life Payors: - Reduces long-term costs, co-morbidity costs, and unnecessary surgical costs. Providers: - Allows for screening more patients - Reduces the number of patients lost to attrition - Creates greater demand for necessary, profitable surgical procedures	Customer Relationships  Parents: Provide an app interface with health information. Create a customer community Clinics: Create efficient feedback loops for iterating on the device and deploying new features Insurance Companies: Payer relations team negotiating contracts Channels  Pediatrician: Recommends this patch to parents as an alternative to a clinic sleep study Sleep Medicine Doctor: Prescribes this test to a patient to evaluate for ENT surgical need Anesthesiologist: Uses device to monitor surgical recovery	Customer Segments  Parents of at-risk children: Earlier diagnosis enables earlier treatment and better outcomes Pediatric Sleep Clinics: Ability to reach all the patients on growing waitlists enables greater throughput and greater revenue Insurance Companies: Inexpensive testing allows payors to avoid covering unnecessary surgeries and decreases long-term comorbidities Surgeons, ENTs, Dentists, PCPs, and Obesity Clinics: Testing for risk management means faster recovery, lower costs, and lower liability
Cost Structure  Research and Development: Ongoing clinical validation, software and hardware refinement Device Manufacturing: Sensing hardware fabrication, assembly, and compliance verification Technology Infrastructure: Forward-deployed engineering for hospital solutions & integration Salaries and Compensation: Paying employees and advisors Customer Acquisition: Creating relationships with clinics & patients driving adoption	Revenue Streams  Academic Medical Centers: Purchase the device to maximize patient pipeline, decrease risk, and achieve gold-standard care Direct-to-Consumer: Parents can purchase test directly to screen children faster than waiting for a slow clinic Private Practice: ENTs adopt this device to screen patients and get them into surgery sooner			

In addition to our ongoing clinical study at Stanford Children's, we spoke to 25+ users just this past quarter:

Patients & Caregivers

The current system is broken for families. They need a simple, at-home option.

"My son is overweight and his doctor said he should get screened for sleep apnea... but I have to take a whole shift off work and drive two hours to the hospital, and I can't do that with my mother ill at home."

Doctors & Payors

Waitlists are crippling care. Doctors need a reliable triage tool NOW.

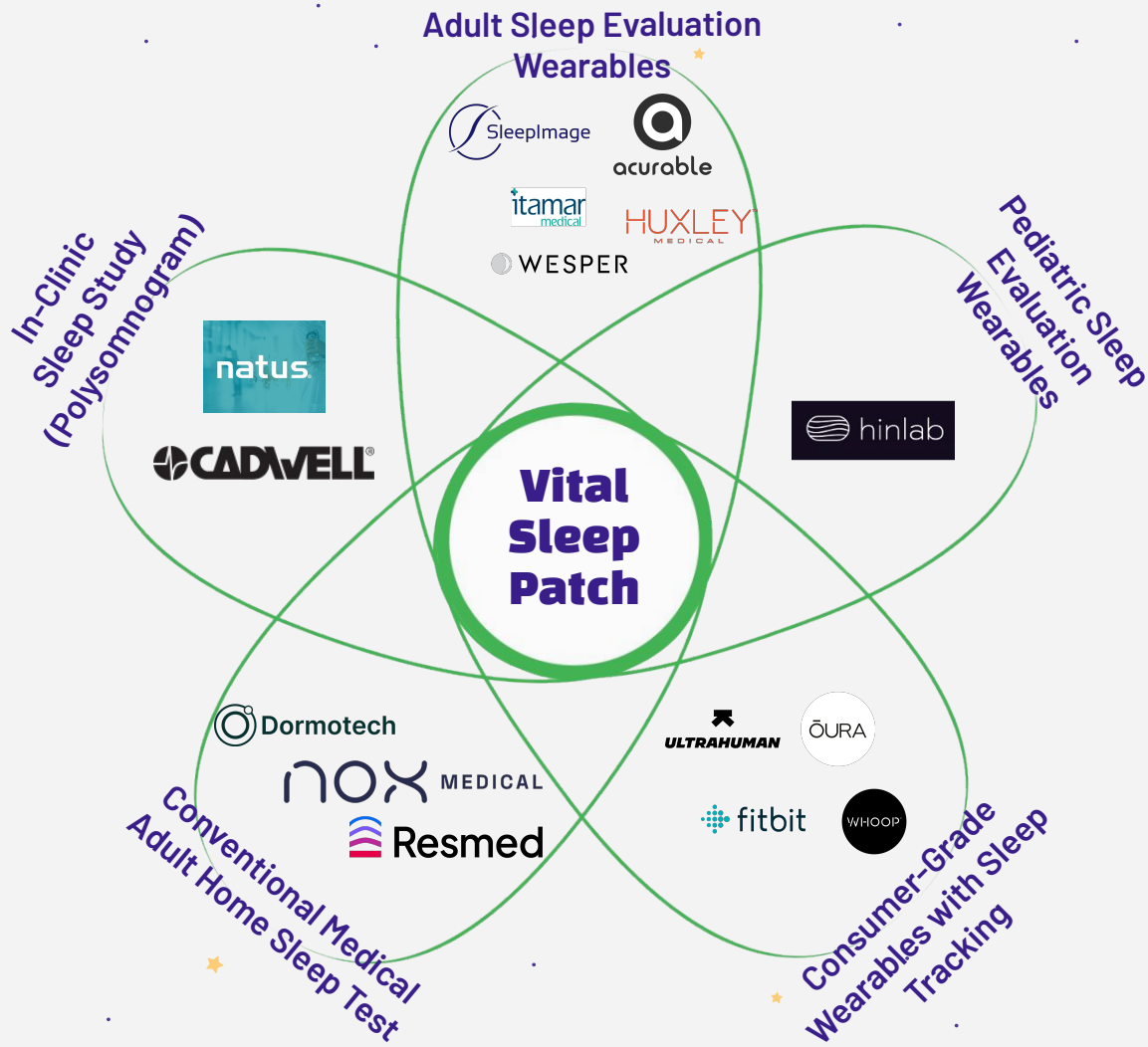
"[I run] a waitlist about 4-6 months out... if we had home devices that worked, [I] would use them 'in a heartbeat' to reach some of the patients that we just don't have capacity to see in clinic"

Wellness Consumers

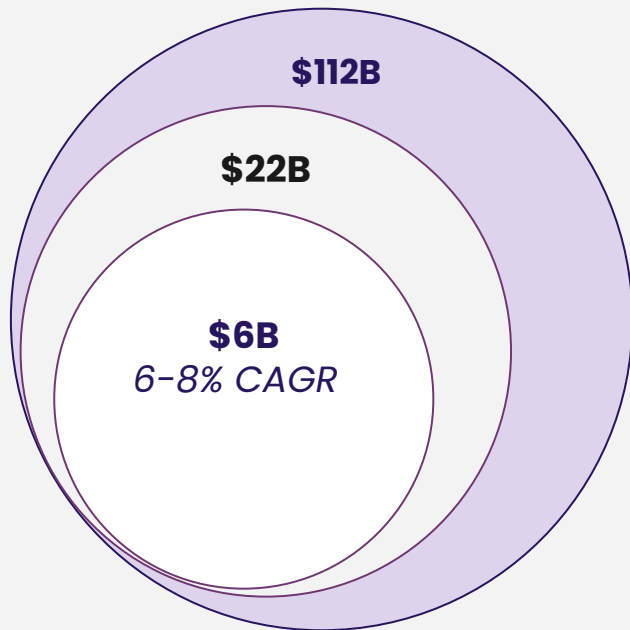
Reliability is the missing link. Prosumers want 'medical-grade' truth.

"I'm willing to spend a premium on something that's actually accurate... I want to feel confident that what I'm being told is real, because that's not always the case."

Competitors



Market Sizing for Vital Sleep Patch



TAM: US Pediatric + Adult Obesity OSA Screenings

SAM: US Pediatric Obesity OSA Screenings

SOM: US Pediatric Surgical Confirmatory Study

Clinical Polysomnogram (PSG): \$10k

Vital Sleep Patch: \$2k

\$300 COGS → 50-80% margin

(SOM) US Peds Suspected OSA: 3MM x \$2k = \$6B

(SAM) US Peds Obesity: 11MM x \$2k = \$22B

(TAM) US Peds + Adult Obesity: 56MM x \$2k = \$112B

Further growth opportunities:



Healthcare
beyond the US



Other medical
applications

Eventually, this device can become an at-home health platform for doctors to monitor multiple clinical indications long-term in high resolution

Potential Next Interviews:

1. Dr. Clete Kushida
 - Director of Stanford Sleep Medicine Center
2. Dr. David Gozal
 - Pediatric Sleep Researcher; University of Missouri
3. Dr. Shyamal Patel
 - Senior Vice President of R&D at Oura
4. Angela Deal
 - author of *My Tired Brain: A Child's Journey to Understanding Sleep Apnea*
5. Dr. Sarah M. Honaker
 - Psychologist for behavioral sleep medicine; Indiana University
6. Aidan Campbell
 - Former Global Product Marketing Manager, ResMed

AI usage attribution: We used ChatGPT and Claude to navigate the distinctions between different fields in the business model canvas, and fine-tuned according to our market research. We also used these tools to identify some potential interviewees, like the author of *My Tired Brain*.